Hydrologic Modeling

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Housekeeping

- HW 7 due tonight
- Quiz at end of next class
- HW 5 should be returned this week, HW 6 next week

Learning Objectives

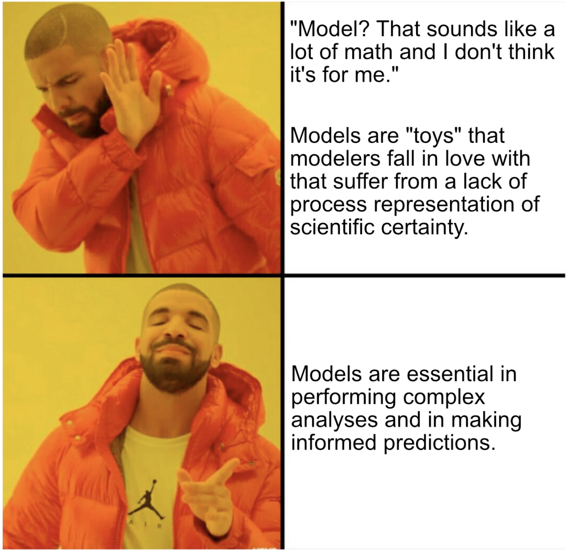
1. Purpose of Models
2. Types of Models
3. Components of Numerical Models
4. Modeling Process
5. Model Evaluation

Hydrologic Modeling

What is a model?

A representation of a portion of the real world "which is simpler than the prototype system and which can reproduce some but not all of the characteristics thereof" (Dooge, 1986).

Opinions of Models



Why Model?

- **Predictive**: Predict hydrologic response to proposed action or future state. Most common, requires calibration.

  "What is the most likely pathway (and transit time) for contaminants if there is a leak in the proposed underground repository at Yucca Mountain?"

- **Interpretive**: Used as framework for studying system dynamics and/or for organizing field data. "Explanatory" modeling. Doesn’t necessarily require calibration.

  "How does groundwater availability and variability influence carbon cycling in the Rio Grande Bosque?"

Classification of Models

- Type: Mathematical - Analog - Physical
- Lumped - Distributed
- Time variant - Time invariant
- Event-based - Continuous
- Static - Dynamic
- Linear - Nonlinear
- Deterministic - Stochastic

Types of Mathematical Models

- **Physically based**: flows and stores simulated using equations derived from basic physics such as conservation of mass/energy/momentum

  Example:

- **Empirical**: data-driven based on statistical/mathematical relationship between observations of inputs and outputs.

  Example:

- **Statistical**: use statistical moments or (auto)correlation analysis to predict future behavior.

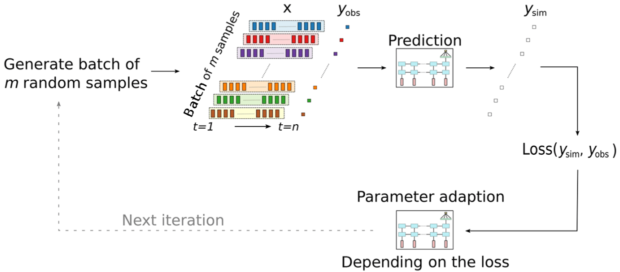
  Example:

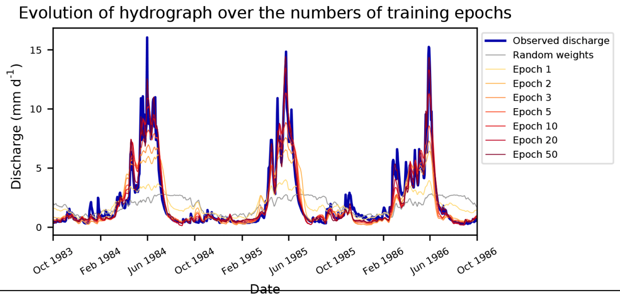
Machine Learning

Data-driven modeling approach that can involve statistical, empirical, and/or physically based components.

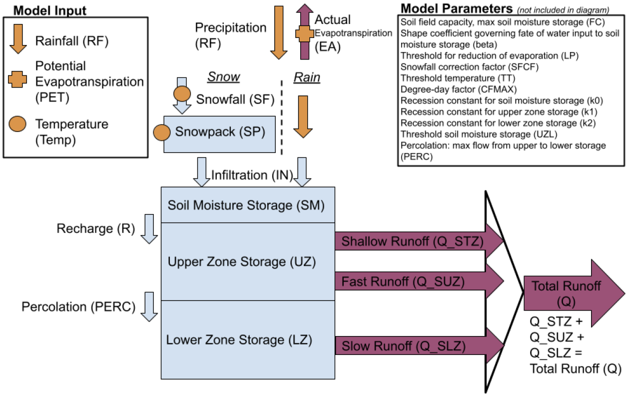
Examples include:

- random forest
- artificial neural networks
- gaussian processes





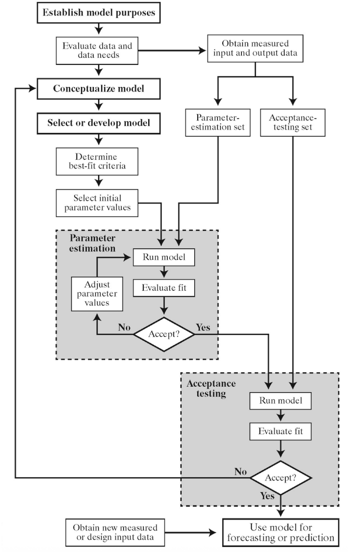
Components of Models

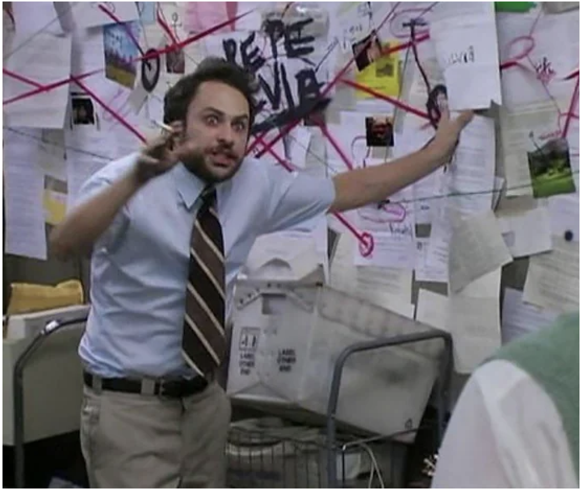


Purple arrows: Output fluxes, Blue Boxes: Storage, Blue Arrows: Fluxes, Orange symbols: Input data

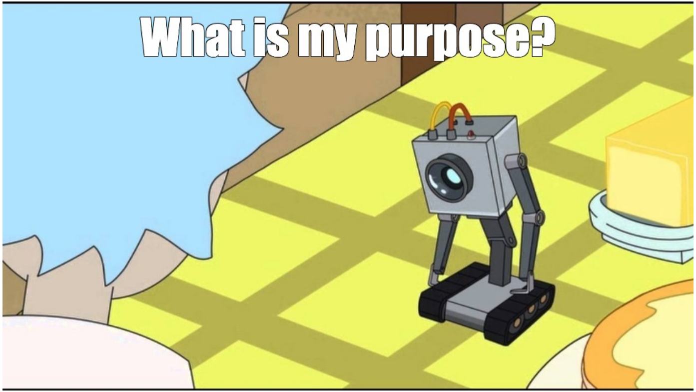
1. Governing equations
2. Conceptual model
3. Inputs
   - Domain (Grid, elevations)
   - Parameters
   - Boundary & Initial conditions
4. Outputs
   - State variables
   - Convergence summary
   - Mass balance

Modeling Process





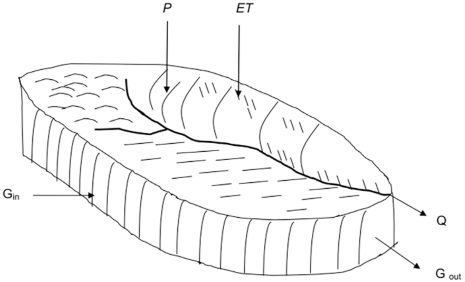
Purpose



- What is the purpose?
- If predictive, what is the prediction of interest?
- If interpretive, what do you want to learn?
- Does this require a numerical model or would an analytical solution suffice?

Conceptual Model

$$\hat{P} + G_{in} - (Q + ET + G_{out}) = \Delta S$$

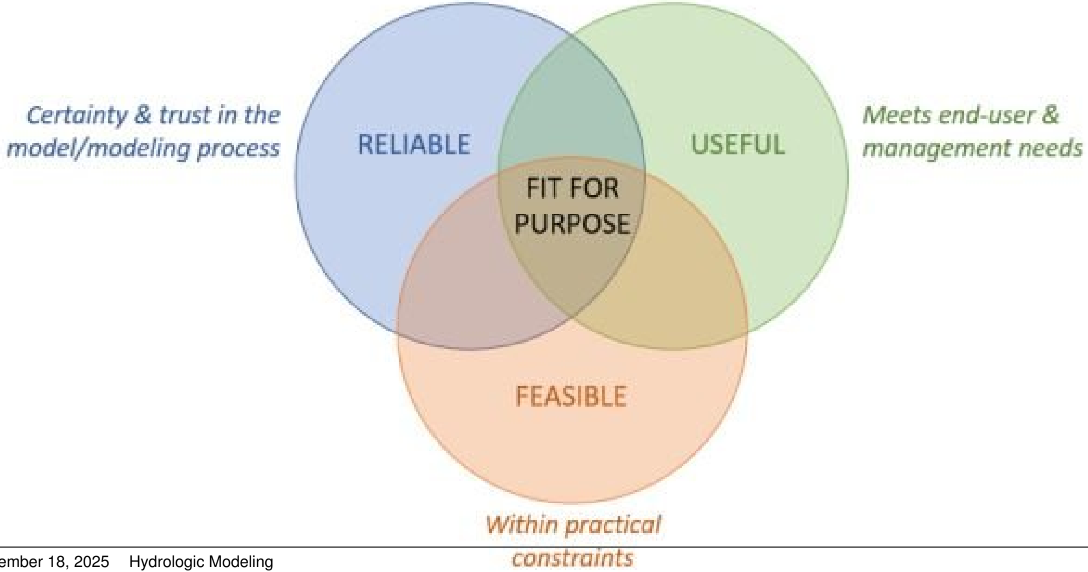


Where:

P	=	Precipitation
ET	=	Evapotranspiration
Q	=	Stream outflow
G_{in}	=	Ground water in
G_{out}	=	Ground water out

- Formulate water balance, hypothesized dynamics, relevant processes
- Gather data
- Review literature

Code Selection



Performance Evaluation

How do you know your model is good enough!?

- Does it reproduce long-term averages in the simulated quantity?
- Does it reproduce spatial/temporal variability in that quantity?
- Is the difference between simulated and observed (residual) sufficiently small?
- Is there spatial or temporal bias in the residuals?

Performance Metrics

Residual:

$$r_i = y_i - x_i$$

Mean absolute error (accuracy):

$$\mu_{|r|} = \frac{1}{N}\sum_{i=1}^{N} |r_i|$$

Mean error (bias):

$$\mu_r = \frac{1}{N}\sum_{i=1}^{N} r_i$$

Root-mean-squared error (accuracy and precision):

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N} r_i^2}$$

Nash-Sutcliffe Efficiency, model as predictor (numerator) versus using mean of the data as predictor (denominator).

$$NSE = 1 - \frac{\sum_{i=1}^{N} |y_i - x_i|^2}{\sum_{i=1}^{N} |\bar{x} - x_i|^2}$$

NSE = 0, model is no better than mean of data. NSE=1 model perfectly reproduces observed data.

where x is observed, y is modeled, N is total number of observation locations and/or times

Build Model

- Design spatial discretization
- Design temporal discretization
- Specify initial and boundary conditions
- Populate spatial units with parameters
- Specify solver parameters
- Run model
- Are the results not obviously wrong?
- Is the model stable and efficient?

Calibration

Calibration, **parameter estimation**, and **history matching** all refer to the process of adjusting model parameters to give the best "fit" to observed quantities based on pre-determined performance criteria.

It is an **inverse problem** because we are using outputs of the model to estimate inputs.

Can be done using **manual**, "trial and error" approach or using **semi-automated inversion**.

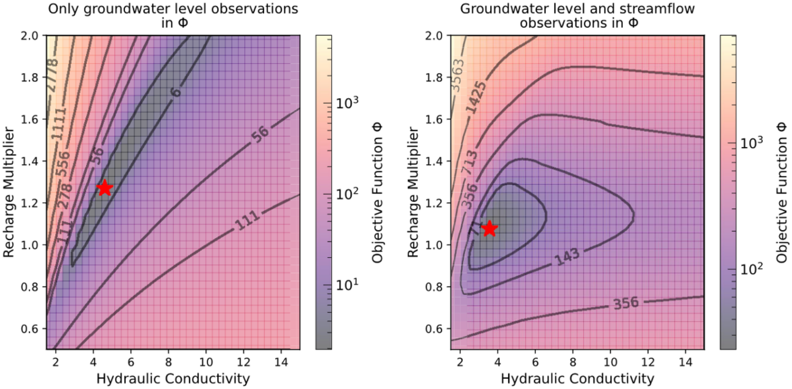
Here Be Dragons

This process is necessary to build confidence in the model but can be fraught due to:

- Equifinality, many combinations of parameter values can yield nearly equivalent fits
- Model simulated quantities may be insensitive to parameters
- Calibrated parameter values may be unrealistic ("parameter compensation)

Equifinality

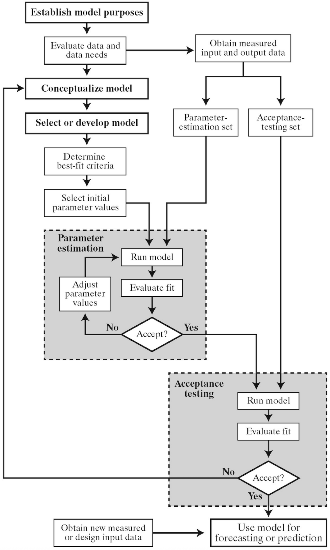
Aka non-uniqueness. Consider a two parameter groundwater flow model (recharge and hydraulic conductivity) with performance metric ϕ.



$$\phi = \sum_{i=1}^{N} (w_i r_i)^2$$

where w is weight (inverse of measurement error). The "best-fit" parameter combination is where ϕ is minimized.

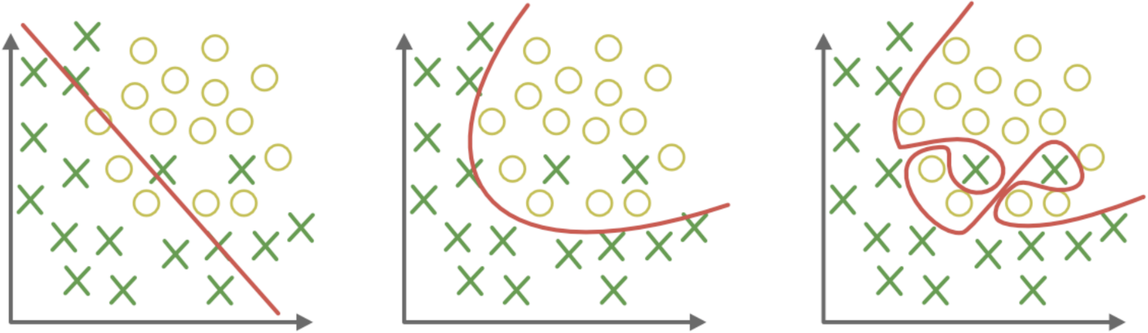
Acceptance Testing



A final step whereby a model is tested against conditions or data that is was not calibrated against to accept or reject it for a particular application.

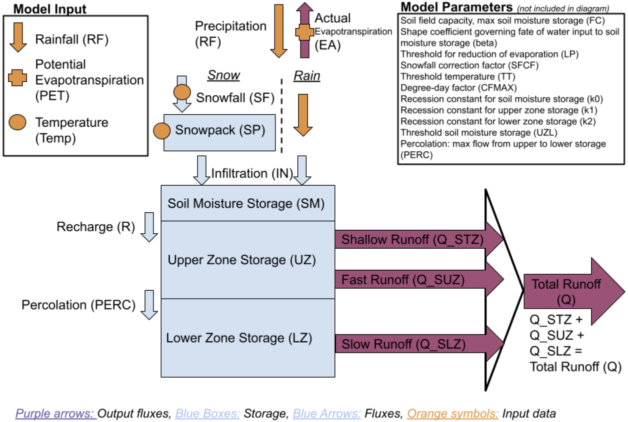
More important for applications in which overfitting can severely degrade predictive power, such as ML and statistical models.

Overfitting



Calibrate your own model!

The HBV model is a semi-distributed model (developed by the Swedish Meteorological and Hydrological Institute) to estimate catchment runoff.

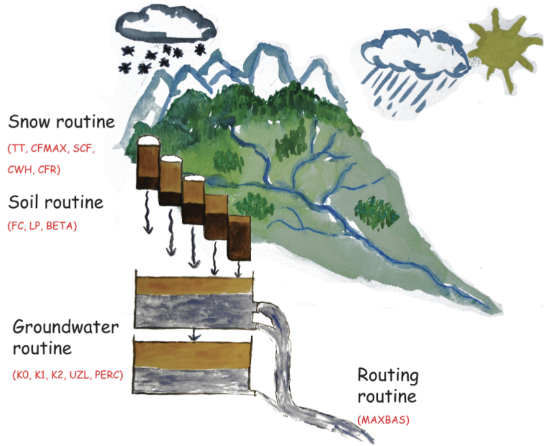


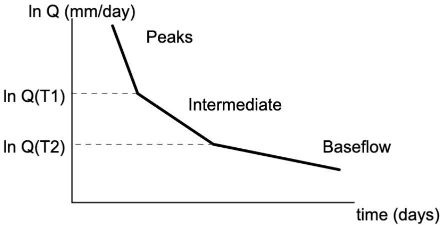
`https://cuahsi.shinyapps.io/HBV-model/`

In-class Activity

1. Get in pairs, we will all agree on the watershed we want to pick.
2. Manually adjust parameters to fit the observed streamflow (using both visual comparison and NSE metric)
3. When finished, click "save this parameter set"
4. Fill in your parameter values and metric to the shared google sheet
5. Click "download this plot" and paste it in the google slide deck

HBV Model





- Slope of the recession:

  Peaks: $K_0 + K_1 + K_2$

  Intermediate: $K_1 + K_2$

  Baseflow: K_2

- Thresholds:

  $Q(T_1) = \sim P_{ERC} + K_1 U_{ZL}$

  $Q(T_2) = \sim P_{ERC}$

HBV Parameter Descriptions

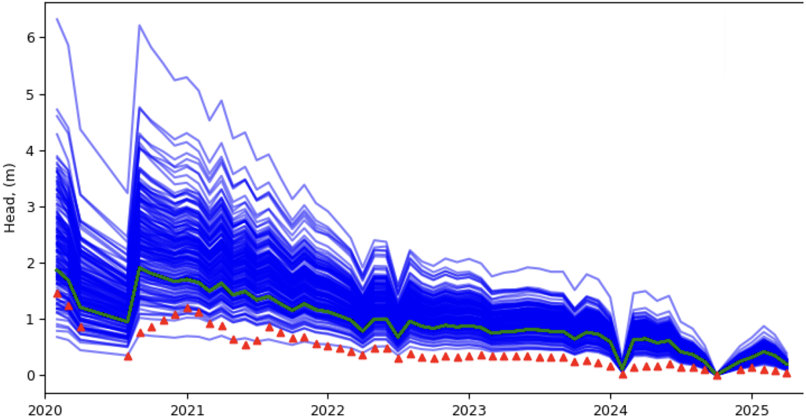
- **Soil field capacity**: maximum soil moisture storage before recharge
- **Shape coefficient for water delivery to soil**: higher value means smaller fraction of precip becomes recharge
- **Threshold for shallow storage**: value that determines whether subsurface runoff is estimate by 2 or 3 linear outflow equations. Smaller = flashier.
- **Recession constant: near surface storage**: slope of recession curve for shallow storage
- **Recession constant: upper storage**: slope of recession curve for middle storage
- **Recession constant: lower storage**: slope of recession curve for deepest storage
- **Percolation: max flow from upper-lower**: higher means more water gets to lower (slower) storage, less flashy
- **Threshold for Evap reduction**: lower value, higher AET

Discussion

- What is an underlying conceptual assumption of the HBV model?
- Did we encounter non-uniqueness? Where?
- What are the challenges of manual calibration?
- Is NSE a good performance metric?
- How could we improve this?

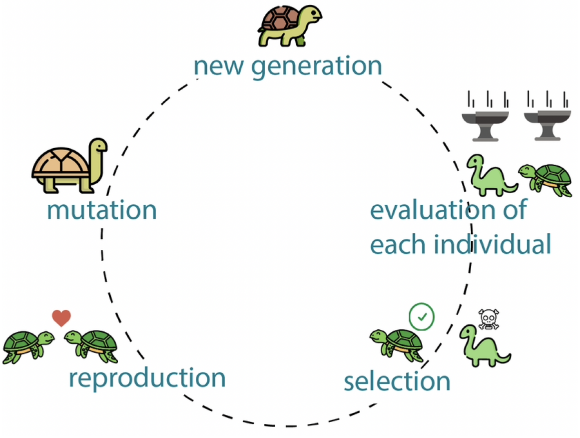
Monte Carlo analysis

Sample from parameter uncertainty ranges to create realizations of models. Run and evaluate.



Genetic Algorithm

Example of semi-automated inversion: optimization approach mimicking natural selection to find optimal combinations of parameters to minimize 1 or more objective functions.



Modeling Myths

- You have to be a math whiz to model
- There's not enough data to build a model
- Complex is always better
- Parsimonious is always better
- A physics-based model is superior to data-driven or empirical